

L12 — GoF Design Patterns

12.1 Creational Patterns

Pattern	Purpose
Singleton	Creates only one instance of a class.
Factory Method	Creates objects through a method.
Abstract Factory	Creates families of related objects.
Builder	Builds complex objects step by step.
Prototype	Creates objects by cloning existing objects.

12.2 Structural Patterns

Pattern	Purpose
Adapter	Makes incompatible interfaces compatible.
Bridge	Separates abstraction from implementation.
Composite	Treats objects and groups uniformly.
Decorator	Adds behavior dynamically.
Facade	Provides a simple interface to a complex system.
Flyweight	Reduces memory usage through object sharing.
Proxy	Controls access to another object.

12.3 Practical — Singleton

```
public class Singleton {  
    private static volatile Singleton instance;
```

```

private Singleton() {}

public static Singleton getInstance() {
    if (instance == null) {
        synchronized (Singleton.class) {
            if (instance == null)
                instance = new Singleton();
        }
    }
    return instance;
}
}

```

Test

```

Singleton s1 = Singleton.getInstance();
Singleton s2 = Singleton.getInstance();
System.out.println(s1 == s2);

```

Output:

true

Adapter Pattern

Existing Class

```

class OldPrinter {

    void printText(String text) {
        System.out.println(text);
    }
}

```

Target Interface

```
interface Printer {  
    void print(String text);  
}
```

Adapter

```
class PrinterAdapter implements Printer {  
    private OldPrinter printer;  
  
    PrinterAdapter(OldPrinter printer) {  
        this.printer = printer;  
    }  
  
    public void print(String text) {  
        printer.printText(text);  
    }  
}
```

Main

```
public class Main {  
    public static void main(String[] args) {  
        Printer p =  
            new PrinterAdapter(new OldPrinter());  
        p.print("Hello World");  
    }  
}
```

Output:

Hello World

Result: Singleton provides a single shared instance, while Adapter makes incompatible interfaces work together.